



PRODUCT NAME

Professional Organic Shower Gel Formula

Description

A cost-effective yet luxurious organic shower gel that combines natural ingredients with a pH-balanced, biodegradable formula. This gel is designed to cleanse effectively while maintaining the skin's natural moisture barrier.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	12 months
Batch Size:	10-100 liters
Processing Time:	2 hours
Temperature:	70°C during production, cool down to 40°C
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessels, Stirrer, pH meter, Heating source
Certification:	EcoCert, COSMOS

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent
2	Cocamidopropyl Betaine	Cocamidopropyl Betaine	10.0%	Mild surfactant and foaming agent
3	Decyl Glucoside	Decyl Glucoside	8.0%	Mild non-ionic surfactant
4	Xanthan Gum	Xanthan Gum	1.0%	Thickening agent
5	Glycerin	Glycerin	5.0%	Humectant
6	Aloe Vera Leaf Juice	Aloe Barbadensis Leaf Juice	5.0%	Moisturizing and soothing agent
7	Citric Acid	Citric Acid	0.5%	pH adjuster
8	Preservative	Sodium Benzoate	0.5%	Preservative
9	Essential Oil (Fragrance)	Lavandula Angustifolia (Lavender) Oil	0.5%	Fragrance and aroma
10	Green Tea Extract	Camellia Sinensis Leaf Extract	5.0%	Antioxidant and skin conditioning agent
11	Hydroxyethylcellulose	Hydroxyethylcellulose	5.0%	Thickening agent
Total			100%	

Manufacturing Process

Phase 1: Aqueous Phase

- Heat water to 70°C.
- Dissolve Xanthan Gum and Hydroxyethylcellulose in water with continuous stirring.

3. Add Glycerin and mix until homogeneous.

Phase 2: Surfactant Phase

1. Combine Cocamidopropyl Betaine and Decyl Glucoside in a separate container.
2. Slowly add to the aqueous phase with gentle stirring.

Phase 3: Cool Down Phase

1. Cool mixture to 40°C.
2. Add Aloe Vera Leaf Juice, Green Tea Extract, and Essential Oil.
3. Adjust pH to 7 using Citric Acid.
4. Add Sodium Benzoate as preservative.

Required Equipment

Mixing vessels, Stirrer, pH meter, Heating source

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: EcoCert, COSMOS

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10-100 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a generous amount to wet skin, lather, and rinse thoroughly. Suitable for daily use.